

GABRIEL LE GUAY

Born august 9th 2001

Swiss - French dual citizenship

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Paris

EDUCATION

2025 / **Graduate researcher** – MASSACHUSETTS INSTITUTE OF TECHNOLOGY (MIT) - Cambridge, US

8 months full time research in the Quantum Nanostructure and Nanofabrication Group (QNN)

- Designed, simulated, and experimentally validated NbN superconducting circuits integrating nTrons, SNSPDs, and modulators. Developed PyTDGL simulations, LTspice models, and PHIDL/GDS layouts; performed cryogenic measurements and device testing
- Realized a reconfigurable superconducting logic gate (XOR/AND/OR) based on nTrons and SNSPDs for Bell state measurement (arXiv preprint in preparation - Associated provisional patent filed (MIT TLO))

2023-2025 / **MSc - FEDERAL POLYTECHNIC INSTITUTE OF ZURICH (ETHZ)** - Zurich, Switzerland

Master in Mechanical Engineering MSc

- Specialized in Nanosystems
- Photonics, semiconductor, quantum dot, sensors, electronics and cleanroom training

2019-2022 / **BSc - FEDERAL POLYTECHNIC INSTITUTE OF LAUSANNE (EPFL)** - Lausanne, Switzerland

Bachelor's degree in Micro-engineering BSc

- CS, electrical engineering and mechanics
- Applied physics and mathematics

RELEVANT WORK EXPERIENCE

C12 Quantum Electronics – *Quantum measurement engineer*

- Develop and operate dilution refrigerator setups to characterize CNT qubits using low-noise cryogenic and RF measurement techniques.
- Analyze device performance to guide qubit optimization and scalable quantum processor development.

Paris, France
2026

ETH Zürich spin-off – *DIAXXO Project developer - 6 months*

- End-to-end development of an automated machine for PCR disposables
- From mechanical design and CAD to prototype assembly and functional testing

Zurich, Switzerland
Spring 2024

ETML - Industrial Intern

- 3-week industrial internship as part of my Bachelor at EPFL
- Hands-on industry experience – Learned and worked about industrial processes and fabrications

Lausanne, Switzerland
Summer 2021

EPFL - Teaching Assistant

- Supporting the professor by leading exercises and class sessions
- 1st year class: 'Mechanical Physics', and 2nd year class: 'Electronics lab'

Lausanne, Switzerland
2020-2022

EXTRACURRICULAR ACTIVITIES

MODERN PENTATHLON - *High level athlete on ministerial list (2017-2019)*

- 12 years practice: France champion in 2012 and 2016
- Participated in Junior world championship in Lisbon (2015-8th) and in Berlin (2019-11th)

PIANO (2014-2019)

- Concours Steinway – awarded honors
- Interlochen Center for the Arts – Summer camp (2016) - Two months in Michigan for an intense piano camp.

HUMANITARIAN MISSION (Spring 2023)

- Full Time caregiver in a retirement home in Jerusalem for 5 months. Assisted elderly residents with daily living, provided companionship, and supported healthcare staff in routine care.

LANGUAGES & INTERESTS

French
English
Spanish

Native language
Fluent – C1 level
Intermediate – B1 level

- LTspice, MATLAB & SolidWorks
- Coding (C++, Python and C)
- Tennis, football and wrestling
- Chess (rating 1920 - Lichess)